CAD Project 4

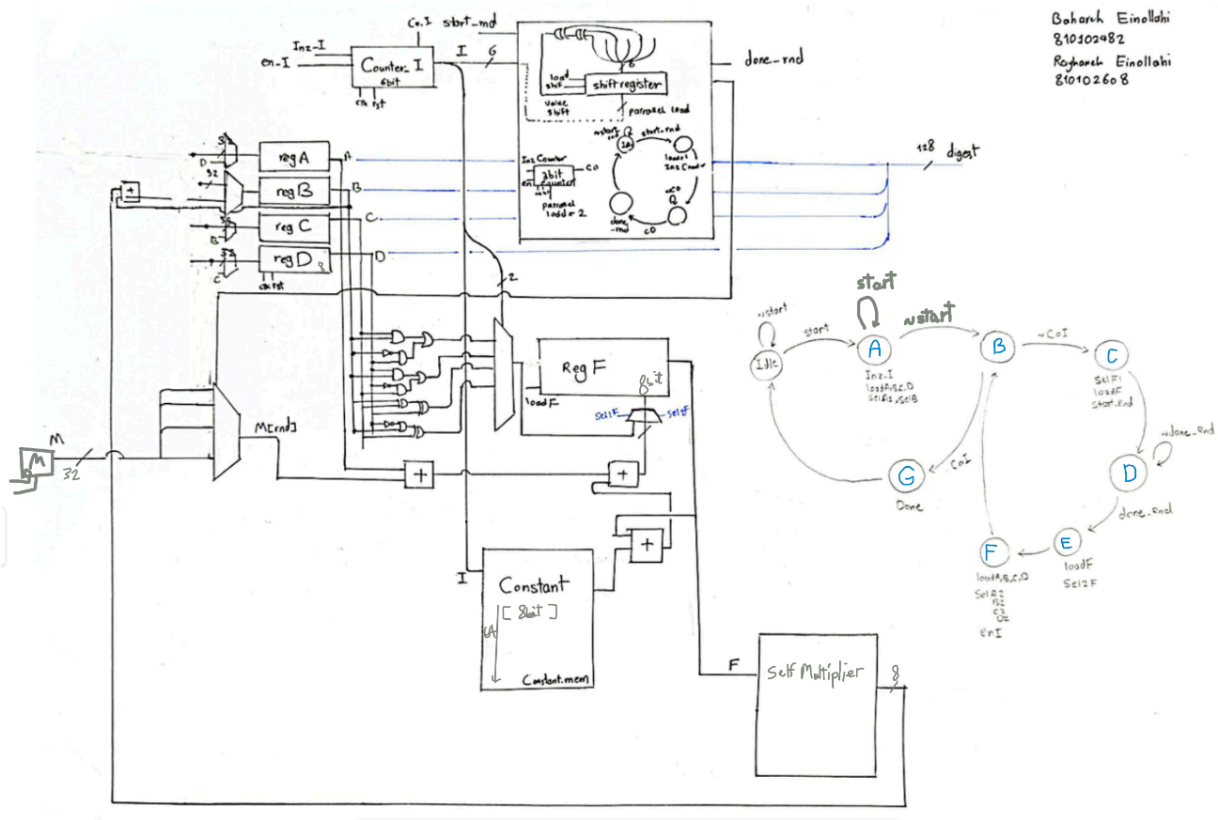
Report and design specification

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بهار 1404

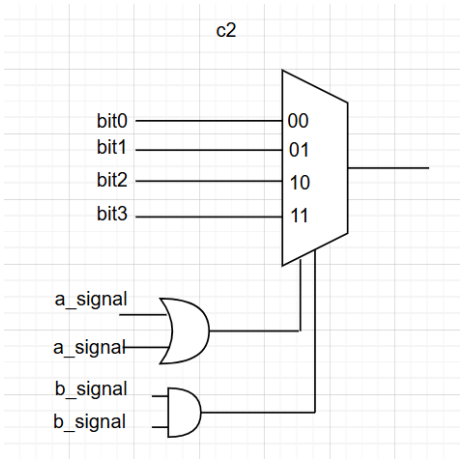
Date path and controller:



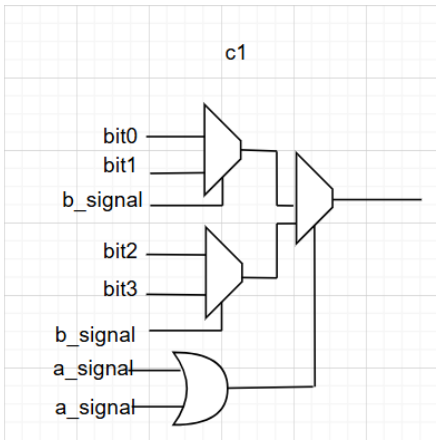
Synthesis details:

```
def map_lut2_to_cell(lut, cell_type, widths) -> Optional[str]:
```

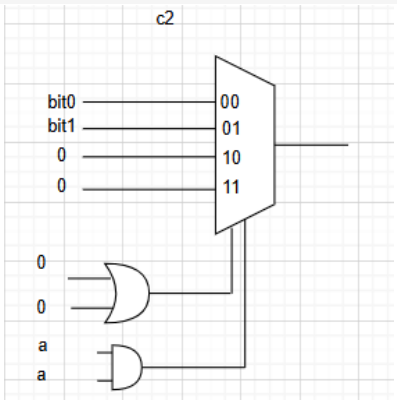
```
if cell_type == "c2":
        return (f"c2 {lut['name']} (\n"
            f"    .D00(1'b{bit0}), .D01(1'b{bit1}), .D10(1'b{bit2}), .D11(1'b{bit3}),\n"
            f"    .A1({a_signal}), .B1({a_signal}), .A0({b_signal}), .B0({b_signal}),\n"
            f"    .out({lut['output']})\n"
            f");")
```



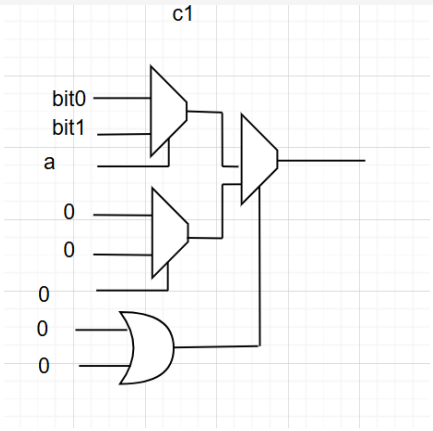

```
return (f"c1 {lut['name']} (\n"
                f"    .A0(1'b{bit0}), .A1(1'b{bit1}), .SA({b_signal}),\n"
                f"    .B0(1'b{bit2}), .B1(1'b{bit3}), .SB({b_signal}),\n"
                f"    .S0({a_signal}), .S1({a_signal}), .f({lut['output']})\n"
                f");")
```



```
if cell_type == "c2":
        return (f"c2 {lut['name']} (\n"
            f"    .D00(1'b{bit0}), .D01(1'b{bit1}), .D10(1'b{0}),
.D11(1'b{0}),\n"
            f"    .A1(1\'b{0}), .B1(1\'b{0}), .A0({a}), .B0({a}),\n"
            f"    .out({lut['output']})\n"
            f");")
```

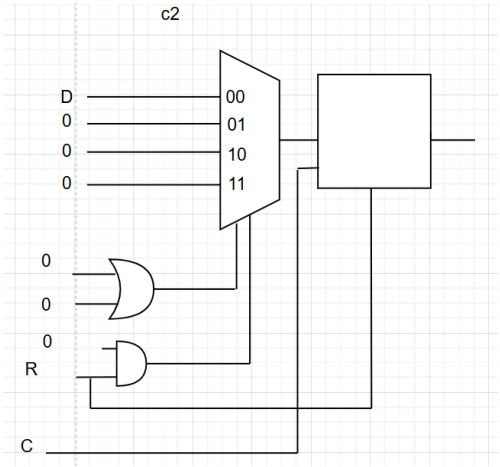



```
return (f"c1 {lut['name']} (\n"
            f"    .A0(1'b{bit0}), .A1(1'b{bit1}), .SA(1'b{a}),\n"
            f"    .B0(1'b{0}), .B1(1'b{0}), .SB(1'b{0}),\n"
            f"    .S0(1\'b{0}), .S1(1\'b{0}), .f({lut['output']})\n"
            f");")
```



```
def map_dff_to_s1(inst: str, ports: Dict[str,str]) -> str:
```

```
return (
        f"s1 {inst} (\n"
        f"    .D00({D}), .D01(1\'b{0}), .D10(1\'b{0}), .D11(1\'b{0}),\n"
        f"    .A1(1\'b{0}), .B1(1\'b{0}), .A0(1\'b{0}),\n"
        f"    .clr({R}), .clk({C}), .out({Q})\n"
        f");"
    )
```




```
def map_sdffe_to_s2(inst_name: str, ports: Dict[str,str]) -> str:
```

```
return (
        f"s2 {inst_name} (\n"
        f"    .D00({Q}), .D01({D}), .D10(1\'b{0}), .D11(1\'b{0}),\n"
        f"    .A1(1\'b{0}), .B1(1\'b{0}), .A0({E}), .B0({E}),\n"
        f"    .clr({R}), .clk({C}), .out({Q})\n"
        f");"
    )
```

ACTEL Mapping Results:
C1:

```
  """Insert a whitespace after any escaped identifier (starting with '\')
Found 0 LUT1 and 892 LUT2 instances (out of 892 $lut)
Total number of c1: 892
Total number of S1: 5
Total number of S2: 56
Created mapped design: mapped_mapped_design.v
Created modules file: modules.v
```

numbers.txt:
10129

Number of Gates for project 3 was 3678 which is not as optimized.